

Liangyu WU | Curriculum Vitae

450 Jane Stanford Way, Stanford, CA 94305, U.S. | E-mail: liangyu5@stanford.edu

Education Background

Stanford University

California, U.S.

Ph.D. Student in Physics (Advisor: **Dr. Julia Gonski & Dr. Dong Su**)

Sep. 2024-Present

- Research area: Experimental Particle Physics

Shanghai Jiao Tong University

Shanghai, China

Bachelor of Science in Physics, School of Physics and Astronomy

Sep. 2020- Jun. 2024

- Overall GPA: 86.1/100 (A-); Physics-related GPA: 90.1/100 (A)
- Coursework: *Quantum Mechanics, Electrodynamics, Equations of Mathematical Physics, Professional Experiment, Introduction to Nuclear and Particle Physics, Methods of Experimental Nuclear and Particle Physics* (All graded A's)

University of Maryland, College Park

Maryland, U.S.

Visiting International Student

Aug. 2023-Dec. 2023

- Overall GPA: 3.925/4.0

Research Experience

SLAC National Accelerator Laboratory

California, U.S.

Particle Physics and Astrophysics

Jan. 2025-Present

Research Assistant (Supervisor: **Dr. Julia Gonski & Dr. Dong Su & Dr. Ariel Schwartzman**)

– Research on the ATLAS Experiment

- Foundation model development with ATLAS Run 2 proton-proton collision data.
- Advancing LAr timing for primary vertex t_0 reconstruction towards 5D calorimetry in ATLAS experiment.
- GigaBit Cable Receiver (GBCR) ASIC Testing for the HL-LHC ITk upgrade.

– Research on the Future Colliders

- On-chip machine learning in future advanced detectors (dual-readout calorimeters, drift chambers, etc.).
- R&D for embedded FPGA (eFPGA) technology in advanced subsystems for future colliders.
- Exploration of R&D on dSiPM technology for future detectors.

SLAC National Accelerator Laboratory

California, U.S.

Particle Physics and Astrophysics

Sep. 2024-Dec. 2024

Research Assistant (Supervisor: **Prof. Spencer Gessner**)

– Research on the FACET-II

- Enhancements to the X-Band TCAV GUI and associated analysis tools.

University of Maryland, College Park

Maryland, U.S.

Department of Physics

Aug. 2023-Aug. 2024

Undergraduate Research Student (Supervisor: **Prof. Christopher Palmer & Prof. Sarah Eno**)

– Research on the Dual-readout Calorimetry

- Geant4 simulations of single-particle responses for diverse calorimeters.
- Formula derivation which predicts the dual-readout-corrected energy from scintillator and Cherenkov signals alone.

Shanghai Jiao Tong University

School of Physics and Astronomy

Undergraduate Research Assistant (Supervisor: **Prof. Yue Meng**)

– **R&D for a Novel Radon Detector**

- Designed the Radic detector and determined the radon diffusion coefficient of several membrane materials.

PandaX Collaboration

Undergraduate Research Student

Shanghai & Sichuan, China

Oct. 2021-Feb. 2023

– **Work in Ultra-low background technique R&D group**

- Measured radon emanation rates across diverse materials for the PandaX and JUNO experiments.
- Conducted ultra-low radioactive surface treatments of materials for TPC assembly in PandaX-4T experiment.

– **Work in PMT group**

- Performed PMT high-voltage testing to ensure standard operating currents and assessed parameters.
- Participated in the installation of veto PMTs and high-voltage cables for TPC-bottom PMTs.

Tsung-Dao Lee Institute

Astronomy and Astrophysics Division

Undergraduate Research Student (Supervisor: **Prof. Masahiro Ogihara**)

Shanghai, China

Mar. 2023-Aug. 2023

– **Research on the stability of unevenly spaced planetary systems**

- Conducted N -body simulations to study the planetary system stability.
- Demonstrated that using evenly spaced models can overestimate orbital stability time in natural systems.

Publications

- [1] S. Eno, **L. Wu** *et al.*, On the resolution of dual readout calorimeters, <https://arxiv.org/abs/2501.15329>
- [2] **L. Wu**, L. Si *et al.*, Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials, **JINST**, **20**, no.03, P03031 (2025)
- [3] S.V. Chekanov, S. Eno, S. Magill, C. Palmer, **L. Wu**, Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders, **NIM A**, Volume **1072**, **2025**, 170200, ISSN 0168-9002.
- [4] S. Yang, **L. Wu** *et al.*, The stability of unevenly spaced planetary systems, **ICARUS**, Volume **406**, December **2023**, 115757

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX

Hardware: Ultra-low radioactive technology, High-voltage technology, PCB debugging, PMT testing

Language: English (Fluent), Mandarin (Native)

Extra-Curricular Experience

Student Union of School of Physics and Astronomy

Co-President of the Student Union

Shanghai, China

Dec. 2021-Dec. 2022

– **Spearheaded the execution of events and promotions for the Student Union.**

Selected Honors and Awards

2023 Best Report First Prize, TDLI Astro-Division Winter Camp (5 out of 96 participants)

2022 NetClass Education and Research Fund Scholarship: First Prize, Shanghai Jiao Tong University (Top 5%)

2021 and 2022 Academic Excellent Scholarship, Shanghai Jiao Tong University (Top 10%, twice)

2021 Chun Geng Education Scholarship, Shanghai Jiao Tong University, School of Physics and Astronomy (Top 5%)